

Geography Ch-12 (Must-Do) — The Oceans: Currents, Tides & Salinity

Must-Do / Foundation+Standard | GS-I (Physical Geography) | Sources: GC Leong + Majid Husain
Ocean currents (wind + Coriolis + density), tides (spring/neap), salinity distribution
CA anchor: AMOC slowdown (thermohaline circulation)

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

FOUNDATION — Currents, Tides & Salinity (GC Leong / Majid)

■ NEWS TRIGGER

Ocean circulation drives climate, fisheries and the monsoon — now in the news as the Atlantic's overturning circulation (AMOC) weakens.

■ INTRO & ORIGIN

Surface currents are driven mainly by **prevailing winds**, shaped by the **Coriolis effect** into **gyres**; deep currents are density-driven (temperature + salinity). Tides come from the Moon/Sun's pull.

Warm currents carry equatorial water poleward; cold currents carry polar water equatorward. Their meeting and upwelling zones make the world's great fishing grounds.

■ KEY DATA

Theme	Key point	Example / note
Current cause	Winds + Coriolis + density	Wind is the dominant driver of surface currents
Gyre rotation	Clockwise (N), anticlockwise (S)	Gulf Stream (N); Canaries (E limb)
Warm vs cold	Warm = poleward; cold = equatorward	Gulf Stream (warm); Labrador (cold)
Fishing grounds	Warm-cold confluence + upwelling	Grand Banks (Gulf Stream x Labrador)
Tides	Spring (New/Full) vs Neap (quarter)	Spring = highest range; Neap = lowest
Salinity	High in tropics/enclosed seas	Dead Sea ~high; Baltic ~7 ppt (low)

■ STATIC THEORY

- **Currents** arise from differential heating, prevailing winds, the **Coriolis (Earth's rotation)** deflection, and salinity/temperature (density) differences (GC Leong, Majid).
- **Gyres** rotate **clockwise in the Northern Hemisphere** and anticlockwise in the Southern (e.g. Gulf Stream vs Canaries limbs).
- **Warm currents** (Gulf Stream, Kuroshio) warm poleward coasts; **cold currents** (Labrador, Peru, Benguela) cool and enrich coasts — their confluence/upwelling gives rich fisheries (Grand Banks; Oyashio-Kuroshio fog).
- **Tides: spring tides** (max range) at New & Full Moon (Sun+Moon aligned); **neap tides** (min range) at the quarters (Sun & Moon at right angles).
- **Salinity** is highest in the tropics (high evaporation) and enclosed seas (Mediterranean, Dead Sea) and low where rivers/ice dilute (Baltic ~7 ppt); ocean average ~35 ppt.

■ MUST-KNOW FACTS

1. Wind is the dominant driver of SURFACE currents; deep currents are density-driven.
2. Gyres: clockwise in N hemisphere, anticlockwise in S.
3. Warm current = poleward; cold current = equatorward.

■ UPSC TRAPS

- ✗ Spring tides occur only in the spring season.
- ✓ 'Spring' means the tide 'springs up' at New/Full Moon — any time of year.
- ✗ Northern hemisphere gyres rotate anticlockwise.

4. Spring tide = New/Full Moon (max); Neap = quarter Moon (min).

5. Average ocean salinity ~35 ppt; Dead Sea very high, Baltic ~7 ppt.

6. Best fisheries = warm-cold current confluence + upwelling.

✓ N-hemisphere gyres rotate **CLOCKWISE** (Coriolis); S-hemisphere anticlockwise.

✗ Cold currents make coasts warmer.

✓ Cold currents **COOL** coasts (and often bring fog/aridity, e.g. Benguela-Namib).

■ MAINS ANGLE

Explain the drivers of ocean currents and tides and their control on climate and fisheries, applying them to the monsoon-reversing North Indian Ocean.

■ STUDY LINK

Geography → Physical → Oceanography (GC Leong: Currents/Tides; Majid: Ocean Currents)

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA ANCHOR — AMOC Slowdown (Thermohaline Circulation)

GS-I / GS-III ·
Currents applied
→ Climate
change

■ NEWS TRIGGER

Studies (2024-25) find the Atlantic Meridional Overturning Circulation (AMOC) — the Atlantic's 'conveyor belt' — is weakening, risking major shifts in global monsoons including India's.

■ INTRO & ORIGIN

AMOC is part of the density-driven **thermohaline circulation**: warm salty water flows north, cools, sinks and returns south at depth — melting freshwater is now slowing it.

■ STATIC THEORY

- **Thermohaline circulation** is driven by **temperature + salinity** differences in density; AMOC carries warm water poleward in the Atlantic.
- Cause of slowdown: melting ice adds freshwater, lowering salinity/density and weakening sinking.
- Impact: a weaker AMOC could shift the ITCZ south and **weaken/delay the Indian monsoon** — a food and water risk.

■ MUST-KNOW FACTS

1. AMOC = Atlantic limb of the thermohaline (density) circulation.
2. Weakening driven by freshwater from ice melt (lower salinity).
3. Risk: southward ITCZ shift -> weaker/delayed Indian monsoon.

■ UPSC TRAPS

- ✗ AMOC is a wind-driven surface current.
- ✓ AMOC is largely a **DENSITY** (thermohaline) circulation, driven by temperature & salinity.

■ MAINS ANGLE

Link thermohaline circulation to climate tipping points and Indian-monsoon security.

■ STUDY LINK

Geography → Oceans → Thermohaline Circulation (applied CA)